Mariel Del Rosario

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EDUCATION

University of California, Berkeley | GPA: 3.9/4.0 | Graduated With General Distinction **Berkeley, CA**
Bachelor of Arts in Geography, Highest Honors *May 2026*

SKILLS

Geospatial Technologies: ArcGIS (Pro, Online, StoryMaps), QGIS, Google Earth Pro, Mapbox GL JS, GDAL, PostGIS, Georeferencing, Digitizing, Geoprocessing, Remote Sensing, Cartography & Map Production, Spatial Analysis (Raster & Vector), Coordinate Systems (CRS)
Programming and Scripting: Python (Pandas, NumPy, Matplotlib, Seaborn, GeoPandas, ArcPy), SQL (Spatial Queries, PostGIS), R, JavaScript, HTML5/CSS3, REST APIs
Data Visualization & Analysis: Data Visualization, Shapefiles, GeoJSON, File Geodatabases, Spatial Data Cleaning, Data Quality, Statistical Modeling, Jupyter Notebooks, Git/GitHub, Excel, Google Sheets

PROJECTS

Web Map: Berkeley Heritage Sites (JavaScript, HTML/CSS, Mapbox) **October 2026**
- Built an interactive web map visualizing 90+ Berkeley heritage sites using Mapbox GL JS, with temporal filters enabling users to explore landmarks by construction era (1997–2007)
- Implemented dynamic point clustering, custom map styling, and custom popups to handle high-density landmark data while maintaining map performance and user readability
- Deployed live at marieltdr.github.io/berkeley-heritage with 50+ unique visitors in the first month

MUNI Transit Suitability Analysis (ArcGIS Pro, Spatial Analysis) **December 2025**
- Identified 3 optimal MUNI station locations across San Francisco using weighted overlay analysis of ridership, accessibility, and socioeconomic equity factors in ArcGIS Pro
- Integrated network analysis, Euclidean distance buffers, and spline interpolation across 5 municipal datasets (census tracts, transit lines, zoning, demographics) to model site suitability
- Published methodology and findings as an ArcGIS StoryMap, translating technical spatial analysis into accessible public documentation for transit planners

Native American Health Equity Spatial Analysis (QGIS, Public Data) **May 2025**
- Cleaned and normalized census and public health datasets to analyze socioeconomic health disparities across Riverside County, CA
- Engineered 4 thematic map layers in QGIS visualizing correlations between the Health Equity Index, land use patterns, and localized health outcomes

EXPERIENCE

UC Berkeley Geography Department **Berkeley, CA**
Geospatial Research Assistant for Dr. Tianna Bruno *August 2024 – Present*
- Georeferenced 50+ historic spatial locations into interactive spatial databases using Google My Maps and Google Earth Pro, launched at tinyurl.com/SWTWP with 200+ unique visitors in the first month
- Built standardized spatial data repositories from oral interviews and archival sources, ensuring data quality control for community-facing cartographic tools
- Co-authored and presented spatial environmental justice research at the 2026 AAG Annual Conference
- Co-developed interactive ArcGIS StoryMaps documenting regional environmental injustice and historical geography in a 4-person research team

College of Computing, Data Science, and Society at UC Berkeley **Berkeley, CA**
Lead Teaching Assistant and Senior Mentor for Foundations of Data Science *August 2023 – May 2026*
- Automated backend workflows for 34 staff members using Python (Google Colab) and Google Sheets API integrations
- Co-authored and debugged Python programming assignments (NumPy) for 1,500+ students, ensuring code quality and pedagogical clarity
- Supervised 40+ TAs/tutors on technical pedagogy, Python debugging, and statistical inference

Pilipinx Academic Student Services **Berkeley, CA**
Retention Director and Gender/Sexuality Awareness Coordinator *May 2023 – June 2025*
- Built a custom spatial database in Google Maps to optimize vendor routing and venue selection for 200-attendee events
- Managed a $26,000+ budget and led a 15-person team organizing student retention initiatives